Expense Voucher Management System

Full Stack Developer Internship Assignment

---

1. Introduction

ABC Company currently manages employee expense vouchers manually. Employees fill out expense vouchers, obtain approval from the Director, and submit them to the Accounts Team for reimbursement. This manual process is time-consuming, difficult to track, and lacks transparency.

The objective of this assignment is to develop a web-based **Expense Voucher Management System** that digitizes the complete voucher creation, approval, and tracking process.

The application should allow employees to create expense vouchers, enable the Director to review and approve or reject them, and allow the Accounts Team to monitor all vouchers for payment processing.

This assignment is designed to evaluate a candidate's ability to build a real-world full-stack application, including frontend development, backend development, authentication, authorization, database design, REST APIs, file uploads, validation, and clean software architecture.

---

2. Objective

The application should:

- Replace the manual voucher approval process.
- Allow employees to submit expense reimbursement requests.
- Allow the Director to approve or reject vouchers.
- Allow the Accounts Team to view all vouchers for reimbursement processing.
- Allow employees to track the status of their submitted vouchers.
- Maintain a centralized repository of all vouchers.
- Ensure only authorized users can access relevant information based on their role.

---

3. User Roles

The application must support the following three user roles.

3.1 Employee

Employees are responsible for creating and submitting expense vouchers.

Employees should be able to:

- Login to the application.
- Create a new voucher.
- Save a voucher as Draft.
- Submit a voucher for approval.
- Edit or delete a voucher while it is in Draft status.
- Upload their signature (image upload).
- View only the vouchers created by themselves.
- View complete voucher details.
- Track the current status of their vouchers.

Employees should not be able to:

- View vouchers created by other employees.
- Approve or reject vouchers.
- Modify submitted, approved, or rejected vouchers.

3.2 Director (Admin)

The Director is responsible for approving or rejecting expense vouchers.

Director should be able to:

- Login to the application.
- View all vouchers.
- View pending approval requests.
- Search, filter, and sort vouchers.
- View voucher details.
- Approve vouchers.
- Reject vouchers.

- Upload their signature while approving a voucher.
- Provide a rejection reason while rejecting a voucher.

Director should not be able to:

- Modify the details entered by the employee.

3.3 Accounts Team

The Accounts Team is responsible for monitoring expense vouchers and processing reimbursements.

Accounts Team should be able to:

- Login to the application.
- View all vouchers across the organization.
- Search, filter, and sort vouchers(**Bonus Point**).
- View voucher details.
- View employee and director signatures.
- View approval status.
- Download or print vouchers (optional).

Accounts Team should not be able to:

- Create vouchers.
- Edit vouchers.
- Delete vouchers.
- Approve or reject vouchers.

---

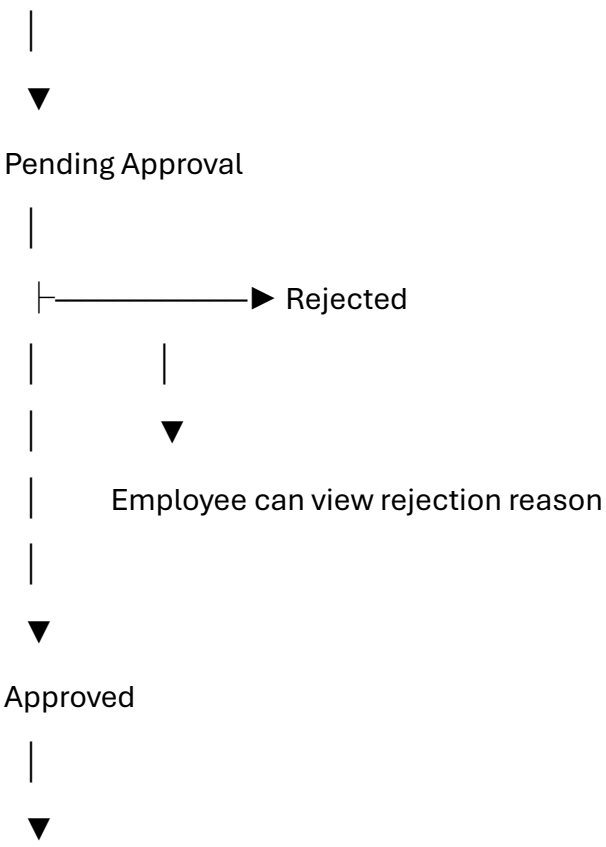
4. Voucher Workflow

Every voucher should follow the workflow below:

Draft



Submitted



---

5. Voucher Details

Each voucher should contain the following information.

Basic Information

- Voucher Number
- Voucher Date
- Expense Date
- Department Name
- Expense Title
- Expense Category
- Expense Description
- Amount

Employee Information

- Employee Name

- Employee ID (Optional)
- Employee Signature (Image Upload)

Approval Information

- Current Status
- Director Signature (Image Upload)
- Approval Date
- Rejection Reason (if rejected)

Audit Information

- Created Date
- Last Updated Date

---

6. Functional Requirements

Authentication

The application shall:

- Allow users to log in securely.
- Restrict access based on user roles.
- Prevent unauthorized access to protected pages.
- Maintain authenticated user sessions.

Employee Module

Employees shall be able to:

- Create vouchers.
- Save vouchers as Draft.
- Submit vouchers for approval.
- Edit Draft vouchers.
- Delete Draft vouchers.
- Upload signature images.
- View all vouchers created by themselves.

- View voucher details.
- Track approval status.

Director Module

The Director shall be able to:

- View all vouchers.
- View pending approvals.
- Search vouchers.
- Filter vouchers.
- Sort vouchers.
- View voucher details.
- Approve vouchers.
- Reject vouchers.
- Upload approval signature.
- Enter rejection remarks.

Accounts Module

The Accounts Team shall be able to:

- View all vouchers.
- Search vouchers.
- Filter vouchers.
- Sort vouchers.
- View voucher details.
- View approval status.
- View employee and director signatures.

---

7. Search & Filter(Bonus Point)

The application should support searching and filtering by:

- Voucher Number
- Employee Name
- Department
- Expense Category
- Status
- Date Range
- Amount Range

---

8. Dashboard

Employee Dashboard

The dashboard should display:

- Total Vouchers
- Draft Vouchers
- Pending Approval
- Approved Vouchers
- Rejected Vouchers
- Total Amount Claimed

Employees should only see vouchers created by themselves.

Director Dashboard

The dashboard should display:

- Pending Approval Count
- Approved Today
- Rejected Today
- Total Pending Amount
- Recent Voucher Activity

The Director should be able to view every voucher in the system.

Accounts Dashboard

The dashboard should display:

- Total Vouchers
- Pending Approval
- Approved Vouchers
- Rejected Vouchers
- Total Approved Expense Amount
- Recent Approved Vouchers

The Accounts Team should be able to view every voucher in the system.

---

9. Validation Rules

The application shall enforce the following validations:

- Department is mandatory.
- Expense Title is mandatory.
- Expense Date is mandatory.
- Amount is mandatory.
- Amount must be greater than zero.
- Employee Signature is mandatory before submission.
- Director Signature is mandatory before approval.
- Rejection Reason is mandatory if a voucher is rejected.

---

10. Business Rules

1. Every voucher must have a unique Voucher Number.
2. Voucher Numbers should be generated automatically.
3. Every new voucher should initially be saved as Draft.
4. Employees may edit or delete vouchers only while they are in Draft status.
5. Once submitted, vouchers become read-only for employees.
6. Only the Director can approve or reject submitted vouchers.

7. Approved vouchers become read-only.
8. Rejected vouchers must include a rejection reason.
9. Employees can only view vouchers created by themselves.
10. The Director can view every voucher in the organization.
11. The Accounts Team can view every voucher in the organization.
12. Every submitted voucher must include the employee's signature.
13. Every approved voucher must include the Director's signature.

---

11. Suggested Screens

Authentication

- Login

Employee

- Dashboard
- Create Voucher
- My Vouchers
- Voucher Details
- Edit Draft Voucher

Director

- Dashboard
- Pending Approvals
- All Vouchers
- Voucher Details

Accounts Team

- Dashboard
- All Vouchers
- Voucher Details

---

12. Technical Requirements

Candidates are free to choose their preferred technology stack.

Minimum expectations:

Frontend

- React (Preferred)
- Responsive Design
- Form Validation
- Protected Routes

Backend

- Node.js
- Express.js or NestJS
- RESTful APIs
- JWT Authentication
- Proper Error Handling

Database

- PostgreSQL or MySQL(can use any other database as well)

File Upload

Support image upload for:

- Employee Signature
- Director Signature

---

13. Submission Requirements

Candidates must submit:

- GitHub repository containing the complete source code.
- README file with:
    - Project setup instructions
    - Database schema explanation
    - API documentation
    - Assumptions made during development

- Database schema or migration files.
- .env.example file.
- Screenshots or a short demo video (optional).

NOTE: - This document and the business requirements contained herein are confidential and proprietary to Prachay Securities Private Limited and are shared solely for the purpose of evaluating candidates for the Full Stack Developer Internship.